CRYSTAL
PRACTICE MANAGEMENT

CRYSTAL PM

HOW-TO GUIDE

Set up a Crystal Meet room

Put a Crystal Meet device in a conference room and join Zoom and Google Meet calls from the room's screen, step by step.

This guide takes one conference room from an empty Raspberry Pi to a room that shows up in the Crystal Meet dashboard and joins calls from its own web page. Plan about 45 minutes per room, most of it waiting for downloads. No programming experience is needed: if you can follow a recipe and type the commands printed here, you can do this. Repeat it for each room.

STEP 1
Prepare the device
Install Raspberry Pi OS, sign in, connect the TV and camera.

STEP 2
Create the room
Add the room on the dashboard's **Provisioning** page and copy its token.

STEP 3
Install Crystal Meet
One command with the room's config file.

STEP 4
First run and screens
Join a test call, pick the TV's screensaver, then point the table screen and the door sign at the device.

BEFORE YOU BEGIN

Hardware per room. Raspberry Pi 4 (4 GB) or Pi 5, the official power supply, a 32 GB microSD card, an HDMI cable to the room's TV, and a USB camera and speakerphone (for example a Logitech Brio 505 and a Jabra Speak2 55).

Two screens on the network. A touch tablet or PoE touch panel on the table to join and control meetings, and a small screen by the door to show whether the room is free. Neither connects to the Pi: each only needs a browser, power, and the office network. Give the room device a fixed address on your router (a DHCP reservation) so the screens can always find it.

The dashboard. The Crystal Meet dashboard must already be running on the office network; the guide "Set up the Crystal Meet dashboard" installs it on its own Raspberry Pi with a fixed address, for example `http://10.0.0.50`. Rooms talk to it on port `3001`.

Accounts. A dashboard sign-in with the admin role, and the Zoom or Google Meet links you want to test with.

1 Prepare the device

- **Flash the card.** On your computer, open **Raspberry Pi Imager**, choose your Pi model, then **Raspberry Pi OS (64-bit)** with desktop, and your microSD card.
- **Pre-fill the settings.** When Imager offers customisation, set the hostname to something like `crystal-meet-room-1`, the username to `pi` with a password you keep, your Wi-Fi network and your time zone, and turn on SSH. Write the card.
- **Connect and boot.** Put the card in the Pi, plug in the TV over HDMI, the camera and speakerphone over USB, then power. Wait for the desktop to appear on the TV.
- **Turn on automatic sign-in.** Open **Preferences › Raspberry Pi Configuration › System** and set **Auto Login** to on, so the room comes back by itself after a power cut.

You should now see the Raspberry Pi desktop on the room's TV, and the Pi should be on the office network.

2 Create the room in the dashboard

- **Open the dashboard** at its address, for example `http://10.0.0.50` or `http://crystal-meet.local`, and sign in.
- **Go to Provisioning** and fill in the **Room Name** exactly as people call the room (for example `Room 1`) and, if you like, its **Location**. Press **Generate Token**.
- **Copy the token** shown under **Enrollment Token**. It is a long string of letters and numbers; you will paste it into the device's config in the next step.

You should now see the token on the page and, on **Devices**, the room listed as provisioning until its device connects.

> **GOOD TO KNOW**
>
> The token links one device to this room and works once. If a device is ever reinstalled, create a new token for it. Nothing else about the room needs to change.

3 Install Crystal Meet on the device

- **Open a terminal** on the Pi (the black icon in the top bar), or connect over SSH from your computer.
- **Get the software.** Type `sudo apt update && sudo apt install -y git`, then `git clone https://github.com/ben-abeo/croom.to.git`, then `cd croom.to`.
- **Make the room's config.** Type `cp deploy/rooms/room-1.yaml ~/room.yaml` (use `room-2.yaml` or `room-3.yaml` for the other rooms), then `nano ~/room.yaml`.
- **Fill in three values.** The room's **name** and, if people call the place something, its **location** (delete that line to show the name alone); the dashboard **url** as `http://<dashboard address>:3001`; and the **enrollment_token** you copied. Save with **Ctrl+O**, Enter, then **Ctrl+X**.
- **Run the installer.** Type `sudo bash installer/install.sh --config ~/room.yaml`. It installs the browser and the room software; expect 10 to 20 minutes.
- **Start it.** When the installer prints **Crystal Meet installation complete**, type `sudo systemctl start croom`. The service also starts on every boot from now on. If it ever stops, the **Start Crystal Meet** icon on the device's desktop (over VNC) starts it again.

  You should now see the room's screensaver fill the TV: the room's name, its status and the time. The room's TV needs no keyboard or mouse, and nobody has to touch it.

4 First run and the screens

- **Open the room page.** On your laptop or phone on the office network, go to `http://crystal-meet-room-1.local:8080`, or use the device's IP address with `:8080`.
- **Check the dashboard.** On **Devices**, the room should show **online** within a minute.
- **Join a test call.** Paste a Zoom link, or a Zoom meeting ID, into **Join with a link** and press **Join**. A meeting ID, or a Zoom link without its passcode, makes a **Passcode** field appear under the box: type the meeting's passcode there before you press **Join** (on the table screen, tapping the field opens the on-screen keyboard on its digits). The TV shows the meeting joining; the page switches to the meeting screen, which offers **Mute**, **Turn camera off** and **Leave**.
- **Leave.** Press **Leave**, then **Tap again to leave**. The TV returns to the screensaver.
- **Pick the TV's idle look.** Tap **Options**, then **Screensaver** on the room page, and choose **Information** (name, status, next booking and the time), **Quiet** (name and status only), **Brand** (the logo) or **Bounce** (the logo drifting around the screen). The TV changes at once and the choice is remembered.
- **Connect the calendar.** To show real bookings and let people join them, follow the guide **Connect Crystal Meet rooms to Google Calendar** next. Until then the room page works with pasted links only.
- **Connect Zoom.** To join Zoom meetings, including ones hosted by other companies, follow the guide **Connect Crystal Meet rooms to Zoom**. Without it, Zoom turns the room away as an automated guest.

- **Set up the table screen.** On a Raspberry Pi with the touch display, clone the repository as in step 3 and type `sudo bash installer/install-kiosk.sh --url http://<device address>:8080/`. The room page opens full screen at every login with its own on-screen keyboard, and a **Room controls** icon on the desktop reopens it if it is ever closed; run the command again to change the address. After updating Crystal Meet, pull the repository on the table Pi too, run the install-kiosk command again and reboot it so the new scale takes effect. On a tablet, open its kiosk browser (Fully Kiosk Browser on Android, Guided Access with Safari on an iPad), set the start page to `http://<device address>:8080/`, turn on full screen and keep the screen awake.
- **Type on the table screen.** Tap the link box (or the passcode field) and the on-screen keyboard comes up from the bottom of the screen. It has three layers, like an iPad's: letters first, then **123** for numbers and the symbols a link or a passcode needs, then **#+=** for the rest; **ABC** goes back to the letters. Tap the shift key (the up arrow) once for one capital, or twice quickly to lock capitals on, as caps lock does, and a dot shows under its arrow; tap it again to let go. Every key answers the moment a finger lands, so both thumbs can type at once, and holding the delete key keeps deleting. Each key clicks softly, and a letter or symbol shows a larger copy of itself above the key while you press it. **Join** on the keyboard joins the meeting you typed; the key with the down arrow puts the keyboard away, and so does a tap anywhere else on the page. To turn the click off, tap **Options**, then **Key click off** under **Keyboard**; **Key click on** brings it back. The room remembers the choice, for every screen that opens its page.
- **Set up the door sign.** On the door screen, set the start page to `http://<device address>:8080/sign`. It shows the room's name, its status and today's next bookings over a photograph of Austin, with a bar down the left edge and a label under the room's name that are green when the room is free, amber when a meeting is about to start, and red while the room is in use or booked. The section "The door sign" after this step says what else it shows and how to choose its background and turn on football scores.
- **Set up the camera and sound.** Tap **Options**, then **Camera framing** (the TV shows the camera between meetings; a screen left alone returns to the idle screen after ten minutes, and **Back** returns at once), press **Find the stops**, nudge with the arrows and zoom, then **Save as Home**. Frame the table, press **Save**, then a preset; repeat for the whiteboard. Tap a preset to return to it, even in a call. Moves take seconds (**Moving…**); the camera goes Home at every service start. During a call the page shows the meeting screen on its own: **Mute**, **Turn camera off**, **Leave**, the volume, **Camera** for a preset or a nudge, and **Upcoming meetings** to see what is booked next. On that screen, under **Sound**, set the level and try **Mute speaker**; `audio.output_device: HDMI` in the room config (then restart) sends the sound to the TV. **No speaker found** or **No controllable camera found**: the device sees no speaker, or no camera that pans or zooms (the MeetUp does).

  You should now see the room page say the room is free again, the door sign's bar and label green, and the dashboard still showing the device online.

> **TIP**
>
> Zoom links that include their passcode (the part after `?pwd=`) join straight through. Google Meet needs the room signed in once as its own Workspace user; the guide "Let Crystal Meet rooms join Google Meet" covers it.

THE DOOR SCREEN

The door sign

The door sign is the page at `http://<device address>:8080/sign`, which step 4 pointed the door screen at. Everything about it is chosen on the table screen, except **This week**, which is a button on the door sign itself: the door screen needs to be a touch screen to use it.

- **What it shows.** A photograph of Austin fills the screen under a navy overlay. Over it are the practice's name beside the logo, the time and the date, the room's name and where it is, and today's next three bookings, each with its time, its title and, for a video meeting, Zoom or Google Meet. The room's status is shown three ways at once, so it can be read from the hallway: a coloured bar down the left edge, a coloured label under the room's name that says it in words, and a ring round the booking in progress. Green is free (**Free until 1:00 PM**). Amber is the ten minutes before a booking (**Customer call starts in 8 minutes**). Red is in use or booked (**In use until 2:00 PM** while the room is in a meeting, **Booked until 2:00 PM** while a booking is under way). On a screen 560 pixels high or less the label before a booking leaves out the booking's title and says **Starts in 8 minutes**: Today shows the title beside it. The photograph itself is never tinted. A sign that cannot reach the device goes grey and says **Sign not connected**.
- **The practice's name.** The name beside the logo is **Crystal Practice Management**. To show another, type `sudo nano /etc/croom/config.yaml` and look under `room:`. If there is a `company:` line, change the name inside its quotes; if not, add a line under `name:` that begins with two spaces, `company: "Lone Star Eye Care"`, so that it lines up with the lines around it. Save, and type `sudo systemctl restart croom`. A `company: ""` line leaves the logo on its own.
- **Choose the background.** On the table screen tap **Options**, then **Door sign**. Under **Door sign background** are nine buttons: **Austin skyline** (the default), **Capitol**, **UT Tower**, **Pennybacker Bridge**, **Hamilton Pool**, **Texas star**, **Hill Country**, **DKR stadium** and **Plain**, which leaves the navy on its own. The one in use is the blue button. The door sign changes within a few seconds. The choice is kept on the device, so it survives a restart, and every door sign of the room follows it. If the sign is plain navy where a photograph should be, check first that **Plain** is not the one in use. A photograph is loaded when the sign opens, so a first load, or a reload, shows navy for a moment (about a third of a second, a second or more on a slow connection). A photograph that has not loaded after 20 seconds is given up on, and the sign stays navy, with a soft glow in the status colour in a corner, until the page is reloaded or the sign has lost the device and found it again.
- **See the week.** Tap **This week** beside today's bookings. The sign shows the next seven days, today first, each with its name, its date and its bookings (the time, the title and Zoom or Google Meet); a day with nothing booked says **Nothing booked**. The seven days run across the screen, or one under another on a sign standing upright, where a booking's title is cut after one line. On a screen lying down and 700 pixels high or less a booking is its time and one line of title, and on one narrower than 1000 pixels the days are named by three letters (**Wed**). A booking is listed once, on the day it starts, and an all-day booking shows 12:00 AM. The sign never scrolls: where a day has more bookings than fit, it shows the first ones and **+3 more**. The week is read from the device when you open it, so open it again to see a change. **Back**, or thirty seconds without a tap, returns to Today. If the week cannot be read the sign says **The week could not be loaded.** and **Back** still works.

- **Football scores, off by default.** On the table screen tap **Options**, then **Door sign**, and under **Football scores** switch on any of **Texas**, **Michigan**, **Texas A&M** and **Missouri** (the ones that are on are blue). A strip appears along the bottom of the door sign. On its left is the school's last result, with the school's own score first whether it played at home or away (**Texas (#4) 45 – Texas A&M 42**). On its right is the next game, the visiting team first (**Next: 10/19 · Texas (#4) @ Missouri (#9)**; a game at a neutral site reads **vs** in place of **@**). A number in orange is a team's place in the Top 25. With more than one school on, the strip shows one at a time and moves to the next every 8 seconds, with the school's name at the start of the line (**Michigan: Michigan (#12) 31 – Ohio State 24**). Switch every school off and the strip goes. Nothing else on the sign moves when the strip comes or goes, except on a short screen (below). The strip never wraps: a name too long for its line ends in "…", and on a screen narrower than 1100 pixels the last result sits above the next game.
- **Where the scores come from.** The device, not the door screen, asks ESPN's public schedule feed at `site.api.espn.com` for each chosen school every ten minutes, and asks for nothing while every school is off. There is nothing to set up: no account and no key. The device needs to reach `site.api.espn.com` over the internet on HTTPS (port 443); nothing else on the door sign needs the internet. The request names only the school and says nothing about the room or the people in it. ESPN does not document this feed or promise to keep it: it can change or stop, and the strip then keeps showing the last result it read until Crystal PM updates the software.
- **On a short screen.** The label under the room's name, the line under it and the room's location are cut with an ellipsis after two lines, and so is a booking's title (after one line in Today on a screen 560 pixels high or less). A room name that would take more than two lines is set in smaller type, as one of more than 24 characters is, so it is never cut; on a screen 560 pixels high or less, one that would take two lines is too, and one that still takes more than two goes on down 0.1 rem at a time, to 1.3 rem, until it takes two. On a screen of any height a name also goes down, to 1.3 rem at the most, while the status would run off the bottom. At 700 pixels high or less the week leaves out the line under the status label and the room's location and makes the room's name smaller, and at 560 pixels or less each day's date too; with football scores on, at 560 pixels high or less, Today's status leaves out the line under its label too.

Photograph credits

The eight photographs are from Wikimedia Commons. They are resized and shown under the navy overlay, each under its author's licence; `CREDITS.md` in the repository has the file names and links.

Background	Photograph by	Licence
Austin skyline	Oleg Yunakov	CC BY-SA 4.0
Capitol	LoneStarMike	CC BY 3.0
UT Tower	Daderot	CC0
Pennybacker Bridge	Dbtfz	Public domain
Hamilton Pool	jcutrer	CC BY-SA 4.0
Texas star (the Bullock Texas State History Museum)	Michael Barera	CC BY-SA 4.0
Hill Country	Earl Mcgehee	CC BY-SA 3.0
DKR stadium	Brint03	CC BY-SA 3.0

IF SOMETHING IS OFF

Troubleshooting

The room page does not load

On the Pi, type `sudo systemctl status croom`. If it is not running, type `sudo journalctl -u croom -n 50` and look for the first line marked ERROR. Make sure port 8080 is not used by something else on the device.

The device shows offline

From the Pi, type `curl http://<dashboard address>:3001/health`. If nothing comes back, the dashboard is not reachable from the room's network: check the address in `~/room.yaml` and that the dashboard Pi is up (its guide has the checks). If the token was already used, create a new one and run the installer again with the updated config.

The browser opens but never joins

Zoom needs the meeting's passcode: paste the full invitation link, or type the passcode in the **Passcode** field that appears for a meeting ID or a link without one (on a "Couldn't join" card that says "Zoom needs the meeting passcode", press **Join again**). Google Meet needs the room signed in (the guide "Let Crystal Meet rooms join Google Meet"); the room page says so when the sign-in is missing or expired. The room page quotes what Meet showed and the log names a screenshot under `/tmp`; from the Pi, `PLAYWRIGHT_BROWSERS_PATH=/opt/croom/browsers DISPLAY=:0 /opt/croom/venv/bin/croom --check-meet <link>` reports the same (stop the service first: `sudo systemctl stop croom`).

No sound or picture in the call

Check the USB camera and speakerphone are connected before the device starts, then reboot the Pi. The service runs as the signed-in desktop user so it can use the room's screen and audio. From the Pi, `PLAYWRIGHT_BROWSERS_PATH=/opt/croom/browsers DISPLAY=:0 /opt/croom/venv/bin/croom --check-meet <link>` (stop the service first: `sudo systemctl stop croom`) prints "Devices the browser sees" with the microphones, speakers and cameras the browser can use; if the MeetUp is missing from "Devices the browser sees" but `wpctl status` lists it, the browser cannot reach PipeWire: make sure `libpulse0` is installed (the installer adds it) and that the service runs in the desktop user's session.

No browser window opens on the TV

The service waits for the desktop before it starts, so give a freshly booted Pi a minute. If the screensaver still does not appear, type `sudo journalctl -u croom -n 50` and look for "Failed to initialize". The browser is installed under `/opt/croom/browsers`; if that folder is missing, run the installer again.

The screens cannot open the page

Use the device's IP address rather than its `.local` name; many tablets do not resolve those. Check that the screen and the device are on the same network, then open `http://<device address>:8080/api/status` from a laptop to confirm the device answers.

The football strip does not appear

A school you switch on shows within a few seconds. If nothing comes, the device is not getting an answer from ESPN. From the Pi, `curl -s -o /dev/null -w '%{http_code}\n' https://site.api.espn.com/apis/site/v2/sports/football/college-football/teams/251/schedule` prints 200 when it can; if it does not, look at the office network for a firewall or web filter that blocks `site.api.espn.com` on HTTPS. `sudo journalctl -u croom -n 50` names the school and the reason (look for "football schedule"). It says so when a school starts to fail, not at every try; the device tries again after 1, 2 and 5 minutes and then every 10, and the strip comes back by itself when a read works. While ESPN cannot be reached the strip keeps the last result it read, so a game played since can still show as next. The strip shows the last game ESPN lists as final and the next one it lists as scheduled, so between seasons it may show the last result alone, for months, until ESPN lists the new schedule.

A choice is lost when the device restarts

The table screen's choices (the background, the football switches and the screensaver) are saved in `control-settings.json` in the device's data folder, `/var/lib/croom` (a `croom` folder under the desktop user's `.local/share` when the service cannot write there). If the file cannot be written, because the SD card is full or read-only or the folder is not the desktop user's, the choices still work, but they are lost at the next restart until the folder can be written. `sudo journalctl -u croom -n 50` says "Could not save the room settings" and why; `df -h /var/lib/croom` shows whether the card is full and `ls -ld /var/lib/croom` who owns the folder (the installer gives it to the desktop user).

A name too long even at the smallest size

A room name goes down to 1.3 rem and no further. Room names over about 150 characters (five lines of the smallest type on a 7-inch 800 by 480 sign) run off the sign; use a shorter name.

GOOD TO KNOW

Under the hood the device software is still called `croom` : that is the name you will see in service and log commands. Everything people see says Crystal Meet.